

TECHNOLOGY STACK ARCHITECTURE

Team ID: SWTID-2026-3238 | Project Name: PocketSmart AI | Track: AI-ML & Generative AI

1. Technology Selection & Architectural Rationale

Layer / Component	Technology Selected	Engineering Justification
Web Framework	FastAPI (Python 3.11+)	High performance async I/O, native Pydantic validation, auto-generated OpenAPI docs.
Frontend Engine	Jinja2 + Vanilla CSS/JS	Server-rendered HTML, zero complex build tooling, ultra-fast initial load.
Generative AI SDK	Google GenAI SDK (<code>google-genai</code>)	Official Google client library with Gemini 2.5 Flash multimodal vision support.
Database & ORM	SQLite + SQLAlchemy 2.0	Zero-configuration relational database with transactional reliability.
Authentication & Crypto	PyJWT + Passlib (Bcrypt)	Industry-standard salted password hashing and stateless token auth.
Testing Suite	Pytest + HTTPX TestClient	Comprehensive test coverage across all routes, models, and fallback logic.